1SAR refinement review

Update: what changed since 2026-05-05

- Same artefact, re-measured: every prior record left byte-identical
- Headline claim re-graded against calibrated bands, not templates
- Three catalog tasks covered for the first time: T14, T15, T16
- First 1SAR sheet to meet the trust invariant without grandfathering

The refinement was sound. The report was not — and the number it reported belongs to no model the agent produced.

Why re-run at all

The model never changed — the instruments did

The May review was issued before the tolerance-benchmark series. It graded its load-bearing claim with a flat, pre-benchmark rule and read a spread across four oracles as tool disagreement. Both of those are now measured quantities.

When	What changed in the harness	Effect on how 1SAR is graded
May 2026	48-round tolerance series not yet run	flat “< 0.05 R-work/R-free gap” template
Since	Registry §3/§4/§6 bands measured and governed	calibrated $\lvert\Delta\rvert$ bands with named preconditions
Since	gemmi routed through toolchain.py, pinned 0.7.5	mask-radii convention made explicit
Since	Trust invariant enforceable (cutover 2026-08-13)	cctbx-only tasks must be waived or closed
Today	Same artefact, five oracle code paths	re-graded, plus three never-run tasks

No-clobber was structural, not careful: outputs are date-keyed, the artefact zip was extracted read-only into a scratch directory, and all 102 pre-existing files were checksummed before and after the run. Every one is byte-identical. April and May records stand exactly as issued.

The R-free finding, sharpened

Same verdict. Much better attribution.

May said

“R-free is reproducibly ~0.21, not 0.199.” Four code paths, 0.2114–0.2209. Three of four fail the < 0.05 gap criterion.

Now measured

The agent's own model header records FREE R VALUE 0.2114. The oracle recomputes 0.2116 — agreement to 0.0002.

So the defect is

in the write-up, not the refinement. The report says 0.199 in eight places; the model it describes does not.

Where does 0.199 come from? No round the agent ran.

Model in the artefact	R-free
1sar_refined	0.2364
round 2	0.2293
round 3	0.2165
final (shipped)	0.2114
round 5	0.2109
round 6	0.2068
report claim	0.199

Every round model carries its R-free in its own PDB header. Measured across all six, the range is 0.2068–0.2364. The reported 0.199 is not a stale value carried from an earlier round — it is below the best round the agent ever produced, by 0.008.

This matters for how the failure is described. “Understated by ~0.012” implies a measurement disagreement. “Corresponds to no model produced” is a reporting-integrity problem, and a different thing to fix.

Re-graded under the current registry

One verdict flips — because the May gap was mask convention, not code

Claim	May grading	2026-09-07 grading	Verdict
R-free 0.199 vs oracle	flat "< 0.05 gap"; 3 of 4 paths fail	§4 recomputed-reference band $\lvert\Delta\rvert \le 0.01$; measured 0.0126	still fails
gemmi vs PHENIX R-free	0.217 vs 0.2116 (Δ 0.0054), read as disagreement	§4 independent-code-path band $\lvert\Delta R\rvert \le 0.02$, matched radii; measured +0.0020	now passes
Round 6 better than final	Δ 0.0048 by oracle	Δ 0.0046 from artefact headers	reproduces

The dissolved disagreement

May measured gemmi under its default vdw mask on gemmi 0.7.4 and got 0.217. Re-run with radii matched to cctbx as the registry now requires, gemmi gives 0.2136 — a 0.0034 shift. The registry records the default-vdw penalty as median +0.0043, measured over 15 entries. The May number was not wrong; it was taken under the convention the benchmark later showed to be the larger of the two effects.

PHENIX model_vs_data
0.2116

gemmi, matched radii
0.2136

gemmi, default vdw (May)
0.217

agent report
0.199

Three tasks covered for the first time

T14 hydrogen placement · T15 structural classification · T16 interface quality

Task	Metric	Value	Status
T14	confident Asn/Gln/His flip-set conflicts	0 of 14 shared	pass
T14	H atoms added (Richardson reduce)	1386	informational
T15	three-state SS agreement — agent model	0.8802	informational
T15	same metric — deposited reference	0.8646	baseline
T16	DockQ vs deposited assembly	0.9445	pass
T16	CAPRI interface quality class	High	pass
T16	buried surface area, chains A/B	437.8 Å²	informational

T16 applies because 1SAR's asymmetric unit carries two protein chains. The agent's A/B pairing reproduces the deposited assembly at CAPRI High — there is no PHENIX interface scorer, so this task is oracle-only by construction.

T14 compares two genuinely independent H builders (Richardson reduce vs mmtbx.reduce2, add_flip_movers=True). They agree on every confident call. phenix.reduce is the same binary as reduce, so its identical count is a redistribution check, not a second opinion.

T15's agreement number is a property of the fold and the two algorithms, not a quality score. It is only meaningful as a difference — which is why the deposited reference was measured too. The agent model sits 0.0156 above it.

Coverage on the example structure

Where the harness has actually been pointed at 1SAR

Covered in May

- T01 superposition
- T03 X-ray refinement
- T05 geometry
- T06 model-vs-data
- T07 map/model
- T10 ligand/site
- T13 data quality

Covered today

- T14 hydrogen placement
- T15 structural classification
- T16 interface quality

Not applicable

- T09 molecular replacement — the model was already placed
- T11 loop fitting — no chain gaps (A 1–96, B 1–96)

Still uncovered

- T02 per-residue comparison — blocked, see next slide

13 of 17 catalog tasks are now settled on this structure: 10 measured, 2 ruled not applicable with a stated reason, 1 blocked by tooling.

Two findings the May run did not have

One about the model, one about the harness

1 · SO4 A 97 is written as ATOM

The deposition writes the sulfate as HETATM; the agent's final model writes it as ATOM. Not cosmetic: a parser that treats ATOM as polymer counts chain A as 97 residues instead of 96. That happened during this run, with gemmi, before the cause was identified.

Related and also new: the agent placed CA A 98 and NA B 98 — both correctly HETATM, and both absent from the deposition, which contains only the sulfate. May assessed their identity; it did not record that neither is in the reference at all.

2 · T02 has no headless path

T02's catalog PHENIX tool, phenix.structure_comparison, is GUI-only in PHENIX 2.0. Its source carries the comment “launches Stucture Comparision GUI by itself, still requires project”. Run headlessly it exits 1 with no output — and raises a project dialog on the user's desktop.

ProSMART, the CCP4 oracle the catalog names for T02, is not installed here. So T02 is not currently satisfiable on this machine by any route the catalog lists.

What the gate caught, unprompted

The hermetic gate rejected this run's first record because four oracle_tool_ref values did not resolve against ref/catalog.yaml — including a combined “reduce vs mmtbx.reduce2” label. The refs were corrected to registered ids. The underlying gap is real and now filed: mmtbx.reduce2 is a dependency of bench_t14_flip_sets.py but is not a registered tool, so no T14 record can name the builder that produced its second opinion.

Trust invariant: closed on merit

The first 1SAR sheet that needs no grandfathering

The trust model forbids grading PHENIX solely with PHENIX. Since the 2026-08-13 cutover that is enforced: a sheet carrying a cctbx-only task must either waive it explicitly or close it with an independent oracle. Sheets issued before the cutover are grandfathered and listed by name — history is history.

Sheet	Tasks	Gap status	Invariant
QDS_1sar…2026-04-24	T06	open — cctbx only	grandfathered
QDS_1sar…2026-04-26	T06, T13	open — cctbx only	grandfathered
QDS_1sar…2026-04-30	T06, T13	open — cctbx only	grandfathered
QDS_1sar…2026-05-01	T06, T13	open — cctbx only	grandfathered
QDS_1sar…2026-05-04	T06	open — cctbx only	grandfathered
QDS_1sar…2026-05-05	T06	open — cctbx only	grandfathered
QDS_1sar…2026-09-07	T03	closed	satisfied
QDS_1sar…2026-09-07	T14, T15, T16	non-cctbx only	satisfied

Today's sheet carries no open cctbx-only row and needs no waiver: T03 has both a cctbx and a non-cctbx R path; T14, T15 and T16 are non-cctbx throughout.

The harness since May

Why the same artefact grades differently now

48-round tolerance series

Every [template] threshold that mattered here replaced by a measured band with stated preconditions — including the R offset that dissolved the May gemmi gap.

Negative-control track

0 of 22 false verdicts on the control set; agents 21/21; the 2VXN case attributed and stood down rather than quietly dropped.

Gate consolidation

Negative-control headlines, governed thresholds and round-count claims are now data checked by the gate, not prose the gate hopes someone re-read.

Toolchain routing

gemmi pinned at 0.7.5 and routed through toolchain.py — which is how this run knew the May legs had run under 0.7.4, and could say so.

Transfer, CI, licensing

Repository public under CultureBotAI with CI green; BSD-3-Clause code, CC-BY-4.0 docs and records.

Net assessment

What holds, what fails, what is still open

Holds

- ✓ The refinement is a real, large improvement
- ✓ Geometry, waters, disulfides, ion identity defensible
- ✓ A/B interface reproduces the deposition (CAPRI High)
- ✓ Both H builders agree on every confident flip
- ✓ SS agreement at the deposited baseline
- ✓ Oracle R-free reproduces May to four decimals

Fails

- ✗ Reported R-free 0.199 matches no model produced
- ✗ …and no round: best was 0.2068 (round 6)
- ✗ The shipped final is not the best round
- ✗ SO4 written as ATOM, not HETATM

Open

T02 blocked on tooling · mmtbx.reduce2 unregistered in the catalog · the T05 rmsz legs still carry their disclosed gemmi 0.7.4 measurement and have not been re-run under 0.7.5.

The one failing claim is unchanged in verdict and clearer in cause. Five code paths now bracket R-free at 0.2114–0.2136 once mask conventions are matched. The gap is not between oracles, and not between the oracle and the model — it is between the model and the report written about it.